

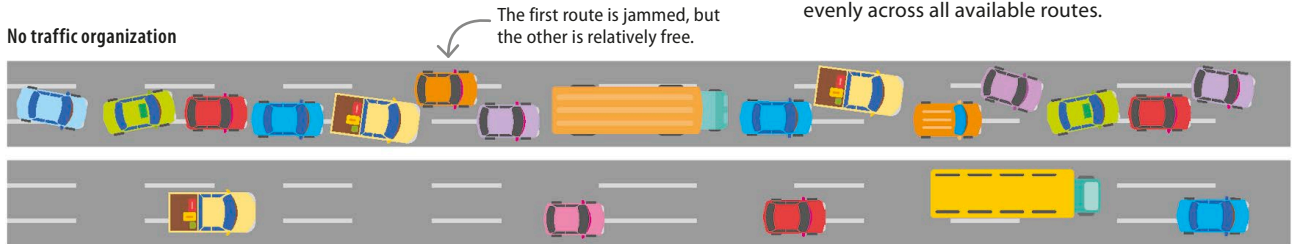
Big data

The IoT has led to extremely large collections of information that are called “big data”. As devices collect more and more information, the data produced can potentially be used to aid humans in real-world situations. Analysing big data to find trends, patterns, and unexpected connections is done by extremely powerful computers.

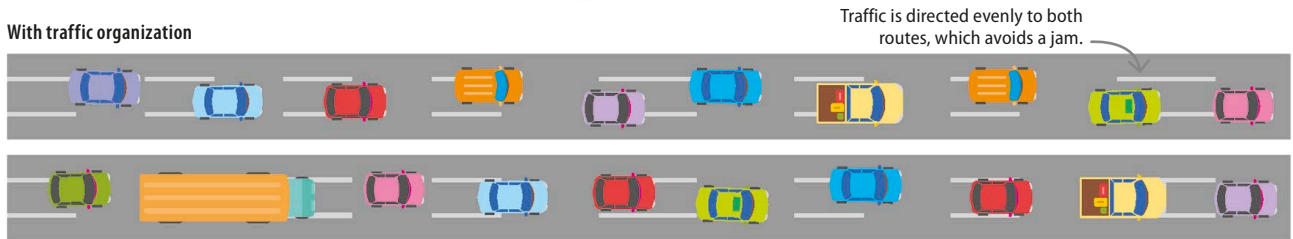
▽ Real-world applications

If nearly every car, bus, and lorry took the same route, then a traffic jam is almost certain, even if another route is relatively free. Big data could potentially help navigation software to automatically direct traffic evenly across all available routes.

No traffic organization



With traffic organization



Privacy issues

While the IoT creates a lot of valuable data, it also creates a lot of data that people might prefer to keep private. Data collected by doctors can save lives, but in the wrong hands, it could be used to target consumer products at patients. Also, any digital system could potentially be hacked, which would make collected data insecure and vulnerable. Installing up-to-date security software, using strong passwords, and practising safe browsing habits prevent most privacy breaches. It's also important to understand how and why data is collected.

“Everything that can be automated will be automated.”
Shoshana Zuboff (b. 1951), professor of business administration, writer



△ Collecting data

Firstly, it's important to know what data is being collected. Smart trackers, for example, can collect data on steps taken in a day, bouts of exercise, and even heart rate and sleep patterns. It's usually possible to request that certain data is not collected in a device's settings panel.



△ Sharing data

Secondly, it's good to know how the data collected is shared. Few people read the terms and conditions when installing apps. Some clauses allow data to be sold to unknown third parties. In the future, data from smart cars and fitness apps could be used to calculate insurance rates.



△ Using data

Finally, it's good to have an idea of how the shared data is used. While shared data from a thermostat or coffee maker might not seem like a big deal, the data can be used to learn a user's habits and create a profile.